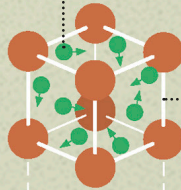




FREE ELECTRON

Early models of solids attempted to model the movement of free electrons as a "Fermi gas" obeying pure Fermi-Dirac statistics.



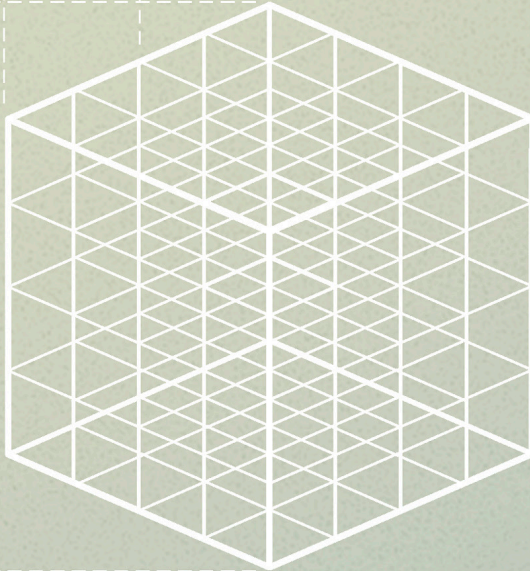
**SECTION
OF LATTICE**



ION

**NEARLY FREE
ELECTRONS**

Modern theories rewrite the wave equation to take interactions between electrons and the positively charged ions of the crystal lattice into account, describing energy levels inside the solid in terms of "electronic bands" (see p.90).



CRYSTALLINE STRUCTURE